

Advanced Computer Networks



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Analysis of 0-persistent CSMA



**DSSRG: Decentralized
Smart Systems Research
Group**

<https://sites.google.com/iitbbs.ac.in/dssrg>

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CSMA 0-persistent

- Analysis only for the **unslotted *non-persistent* CSMA**
- ***Slotted non-persistent* CSMA -- ?**
- **Unslotted *1-persistent* CSMA** protocols will follow this -



Definitions and notations

- **Arrival rate λ = number of *new* messages that arrives per time unit**
- **Offered load (rate) g = number of (old and new) messages that contend for the channel per time unit**
- **Throughput S = number of messages that are *successfully* transmitted per time unit**
- **A message is successfully transmitted if its transmission does *not* collide with any other message transmission.**

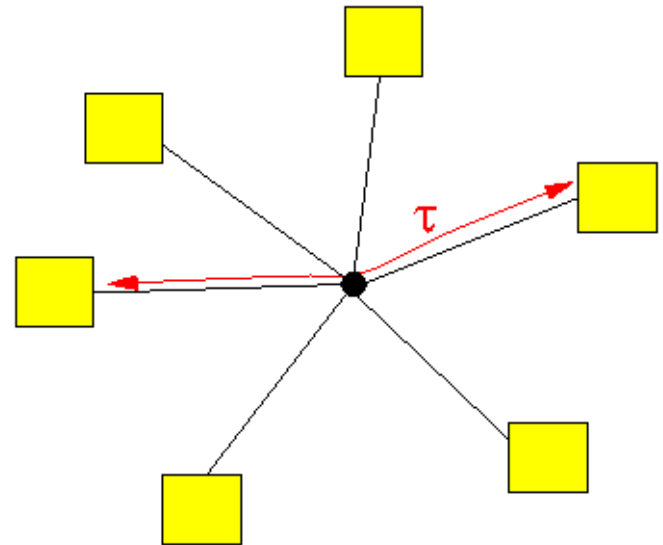


- **Maximum propagation delay τ = the maximum time needed for a transmission from any source to reach all nodes in the CSMA system**
- **The max. prop. delay τ depends on the length of the cables**



Simplifying assumptions

- Simplifying assumptions:
- All packets have the same length T
- (So the transmission time for any packet is the same duration)
- Assume that the propagation delay between *any two nodes* is equal to τ seconds

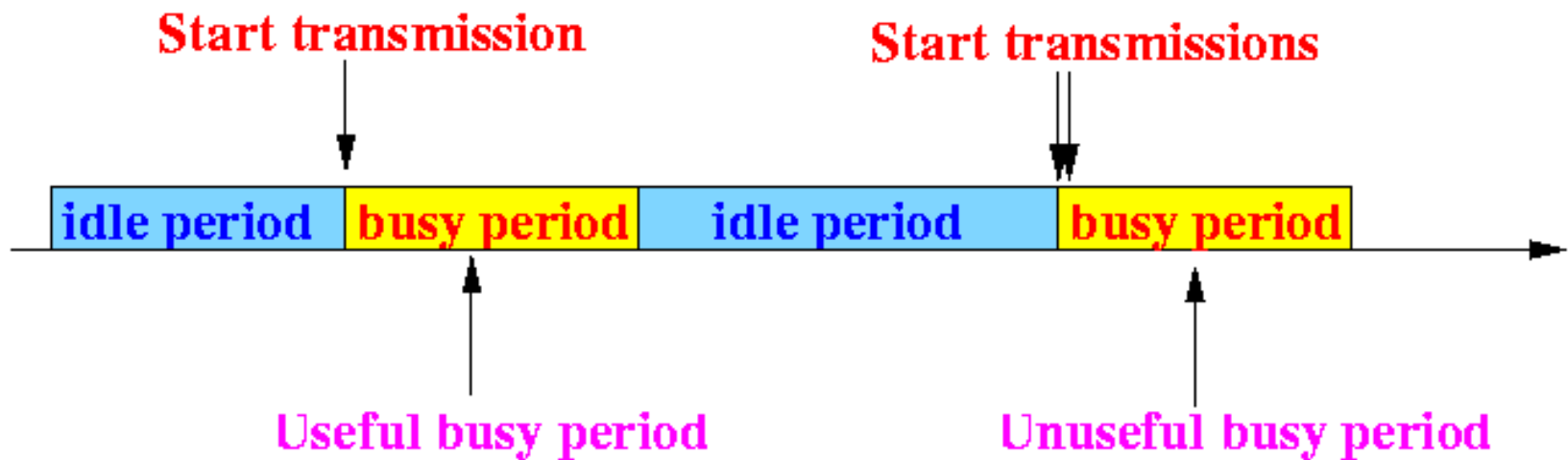


- **Define:** $a = \tau / T = (\text{Propagation Delay} / \text{Transmission Time})$
- *[Think of: One propagation delay is equivalent to how many packet transmissions ? (obviously fraction here)]*
- It is the **normalized propagation delay**
- The **packet arrival process** is a **Poisson process** with rate λ
- The **offered load** (new arrivals and retransmission attempts combined) is **also Poisson distributed** with rate g
(Recall from ALOHA!..)
- (We will see that the **offered load g is dependent on the arrival rate λ**)

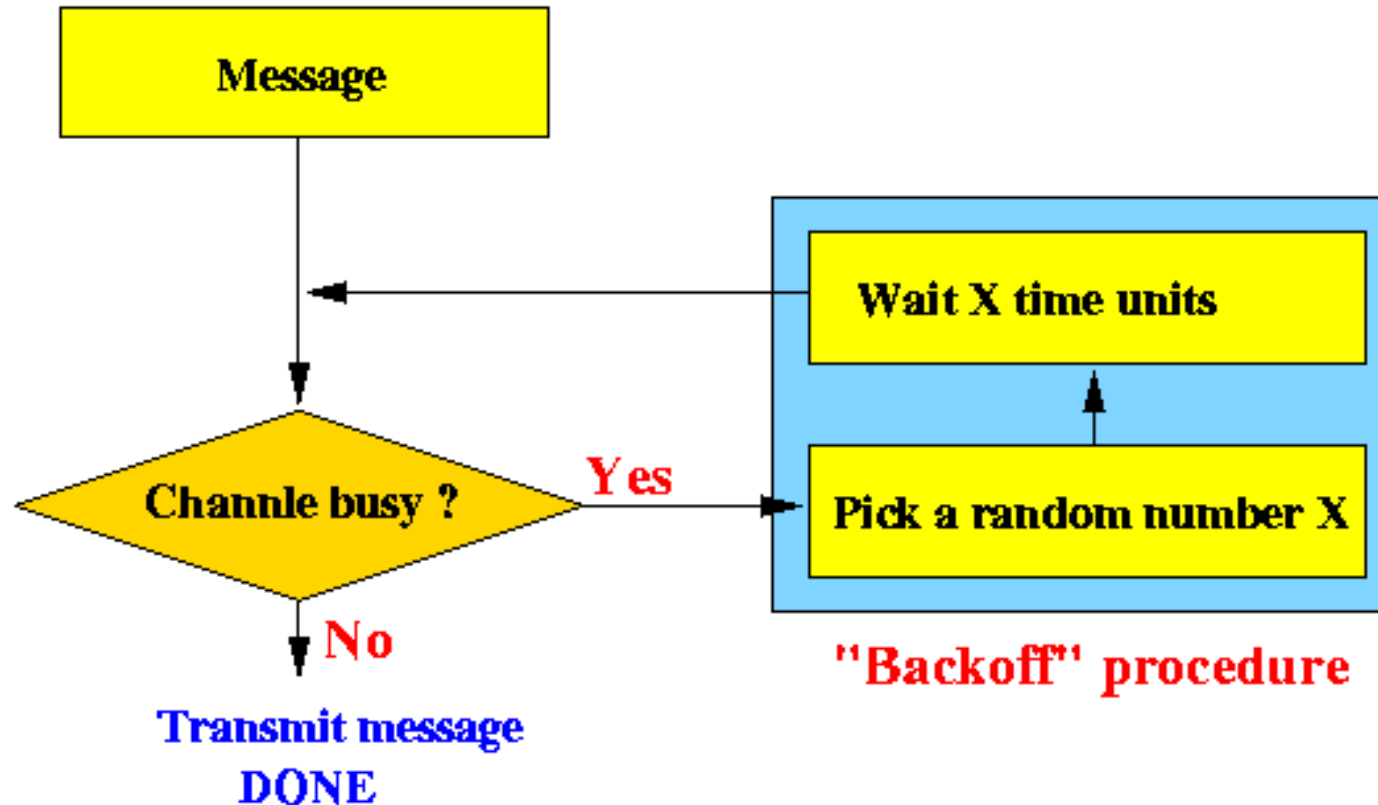


Overview of the system operation

- Abstractly speaking, the *non-persistent* CSMA protocol will cause the **nodes** to transmit in the following manner:
-



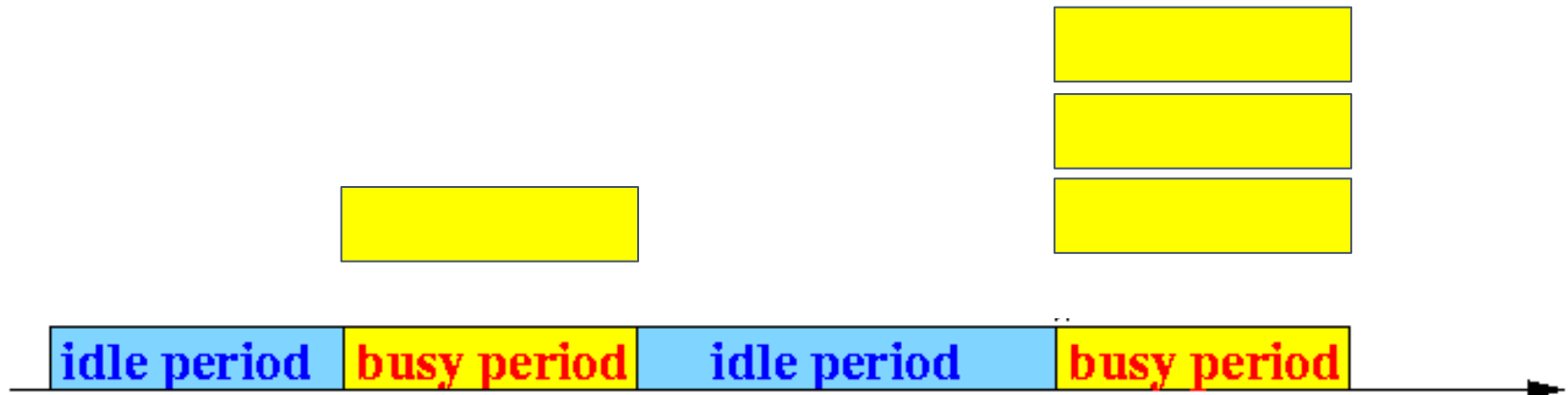
Overview of the system operation



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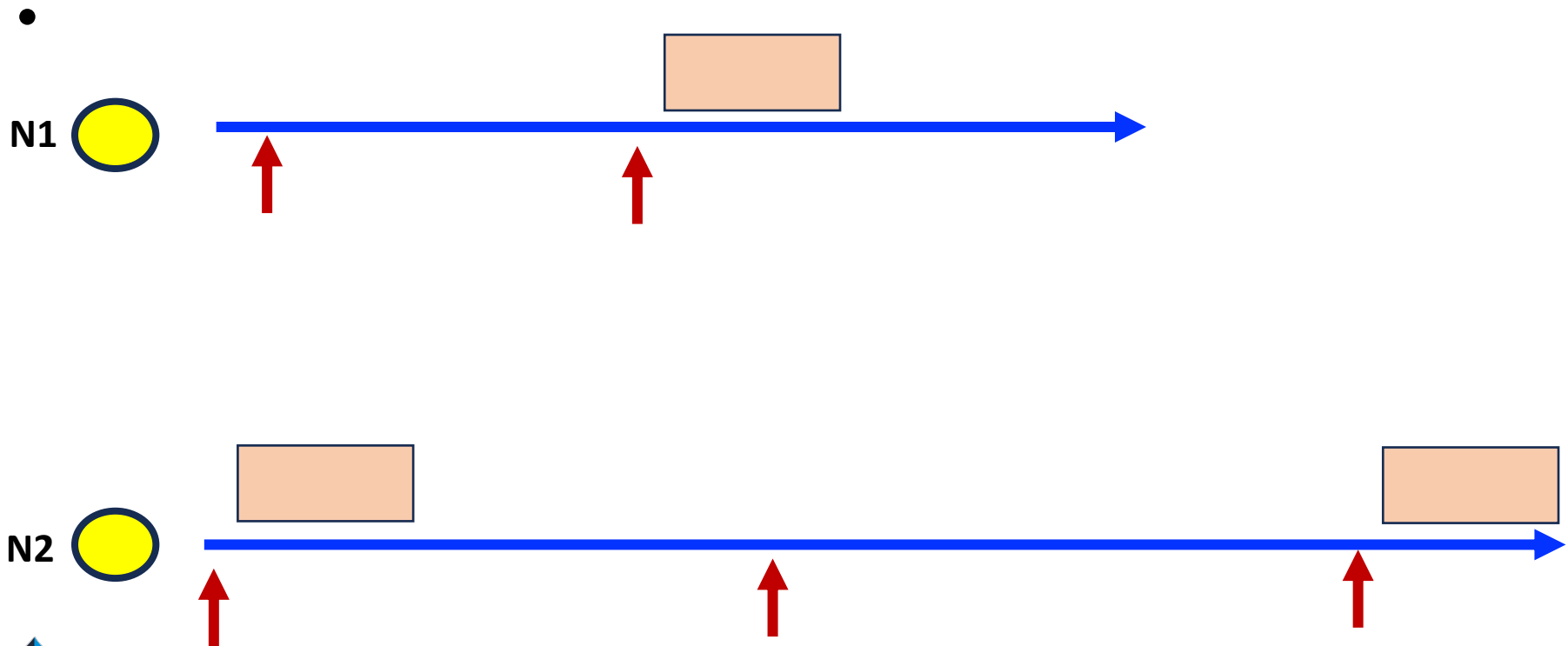
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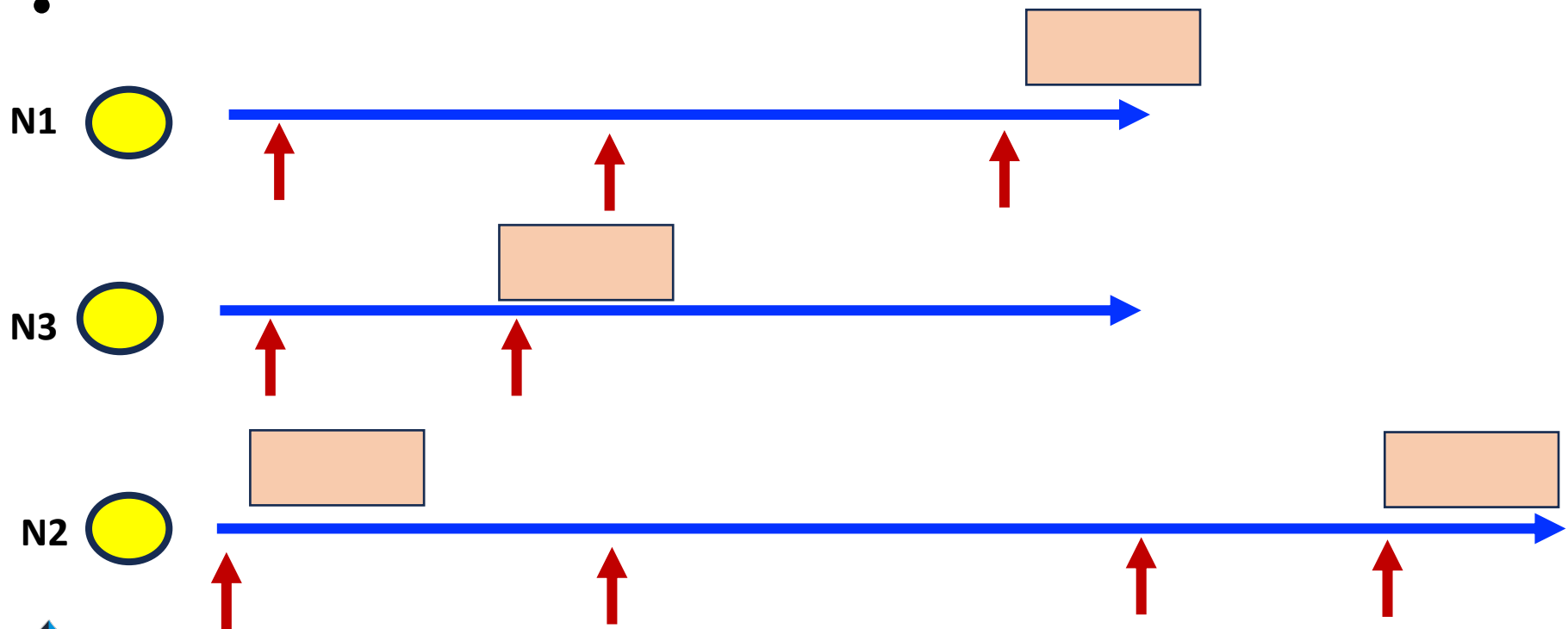
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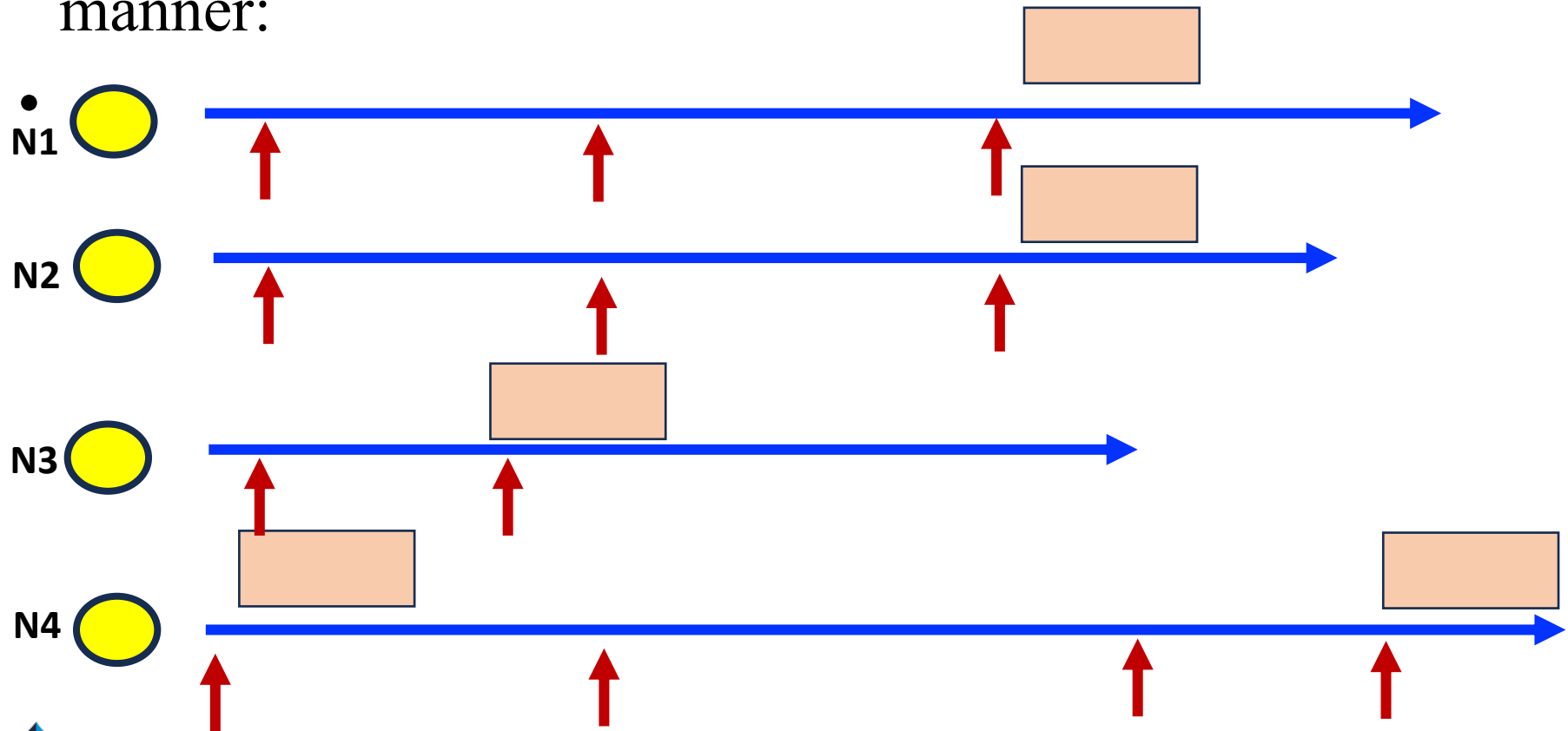
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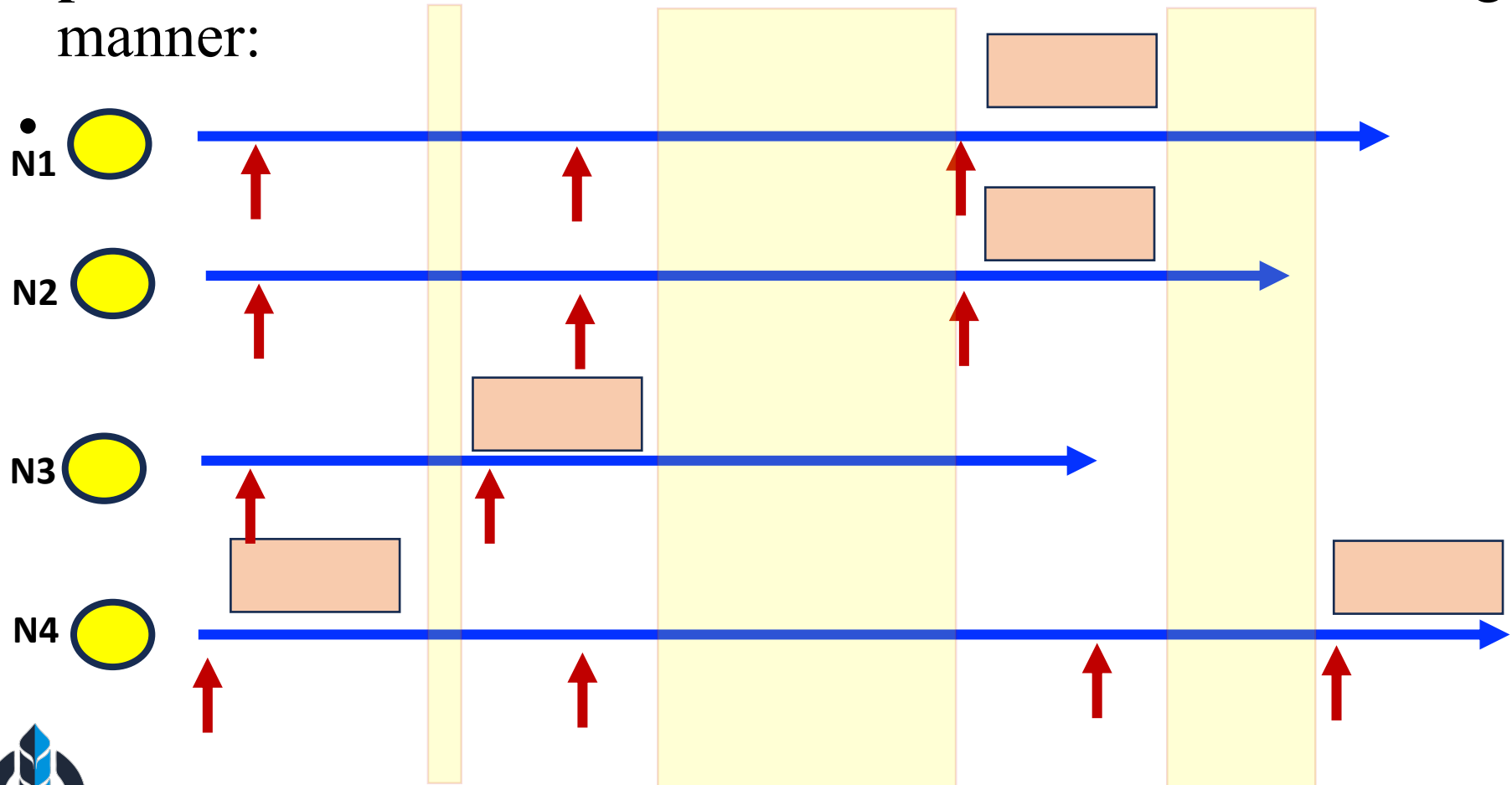
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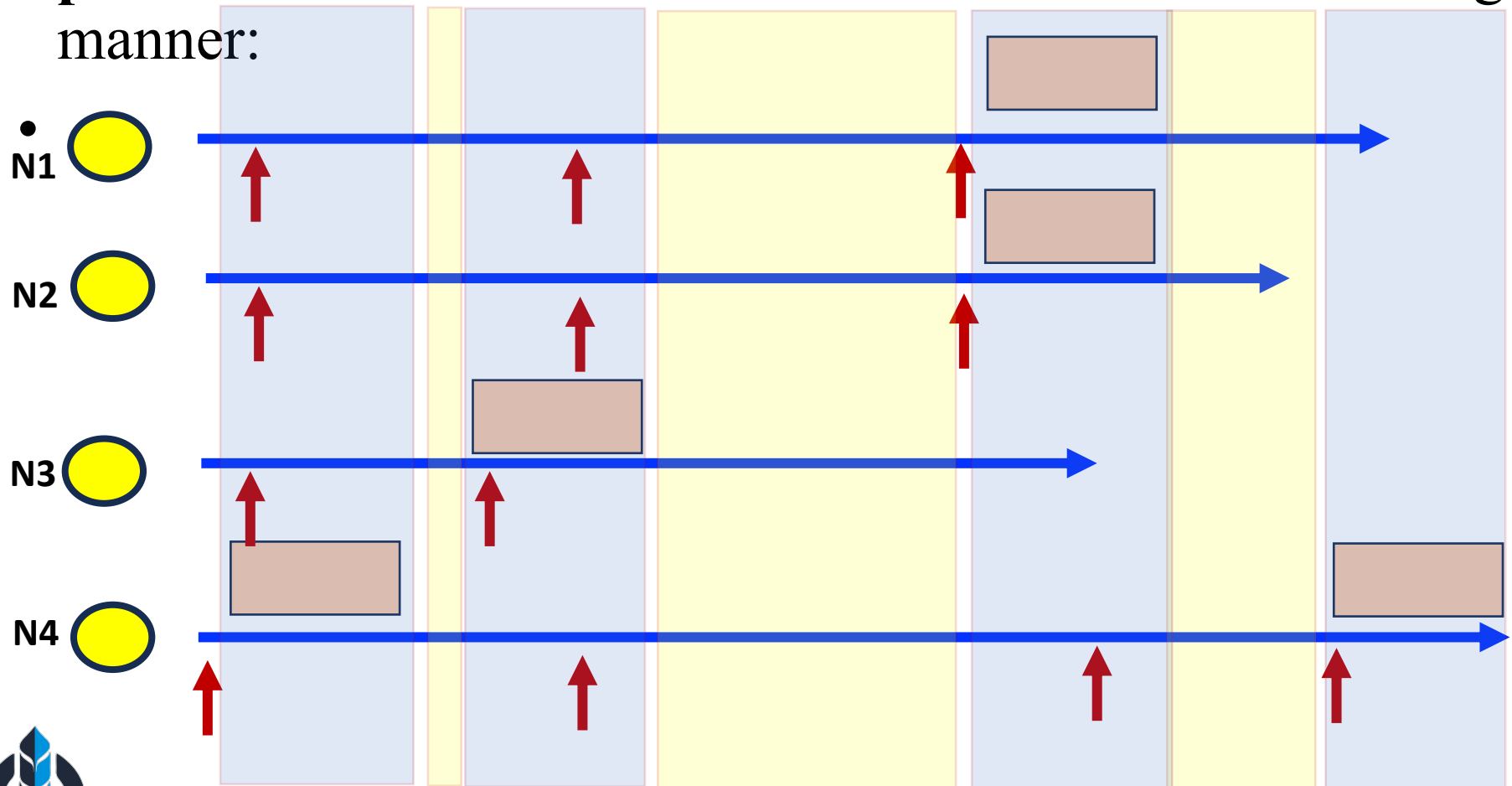
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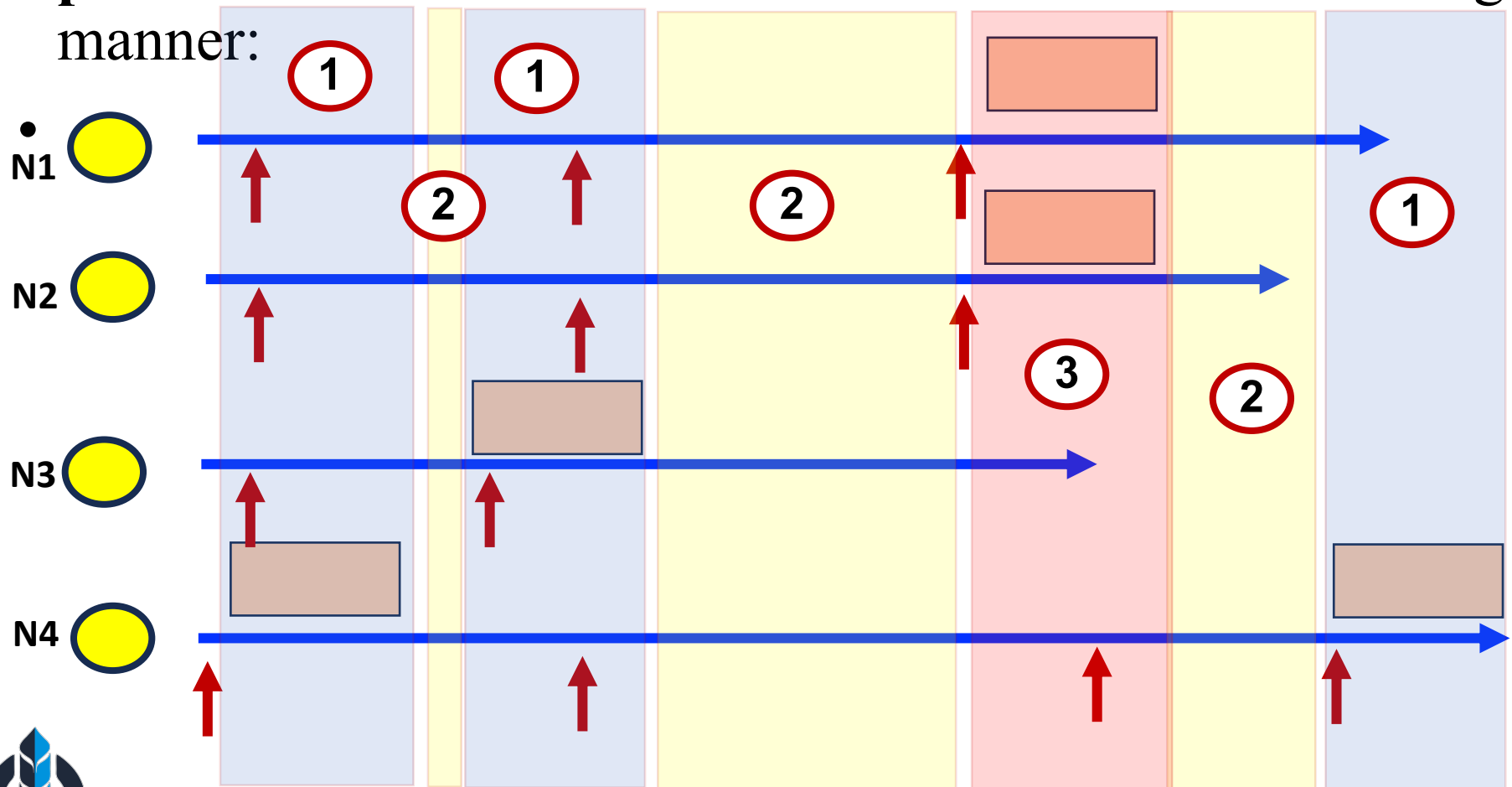
Overview of the system operation

- Abstractly speaking, the *non-persistent CSMA* protocol will cause the **nodes** to transmit in the following manner:



Overview of the system operation

- Abstractly speaking, the *non-persistent CSMA* protocol will cause the nodes to transmit in the following manner:



Notes:

- The **network** is **initially idle**
- Then **one or more nodes** will **transmit**
- If **one node transmits**, then the **busy period** will contain a **successful transmission**
- If *more* than one node transmits, then the **busy period** will contain a **collision (unsuccessful transmission)**
- **Note:**
- Because **CSMA** does **not have *collision detection***, when a **node starts transmitting**, it will **transmit for T sec** (until the packet is transmitted)



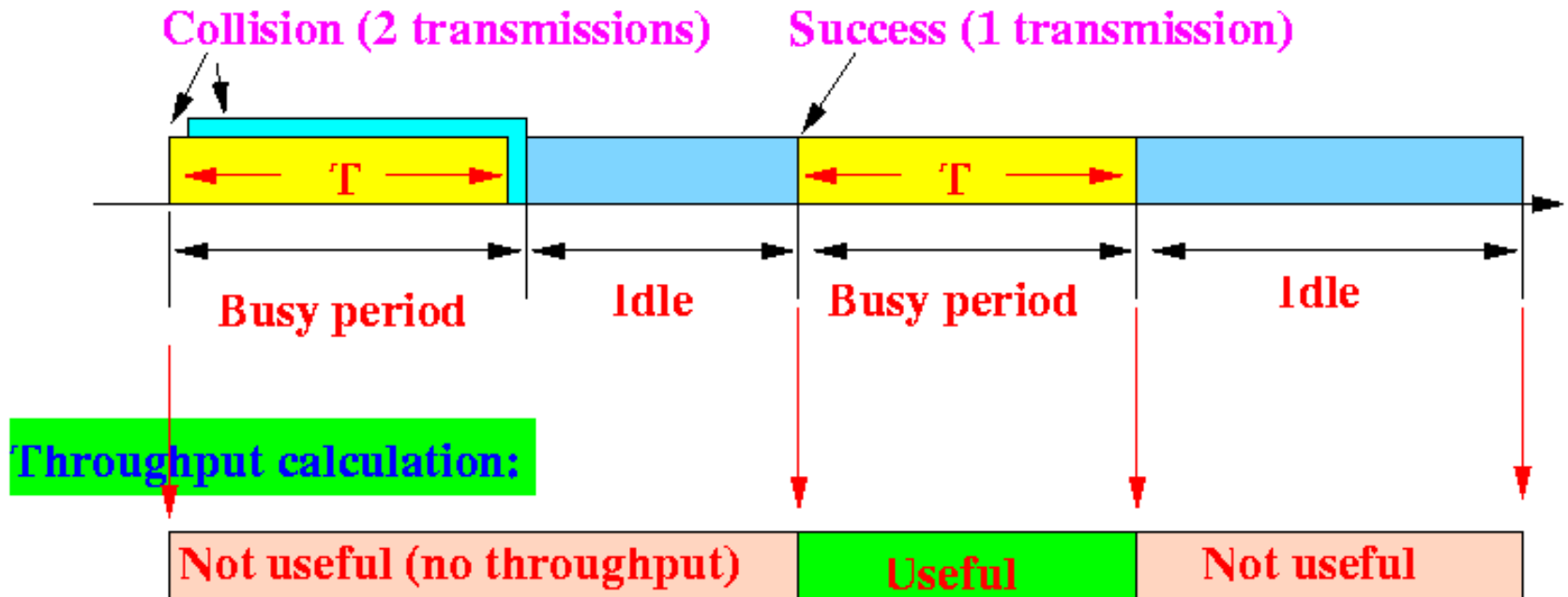
- After the **transmission(s) finishes**, there will be an **idle period** when the nodes will **sense the channel** before they transmit.
- This **idle period** has the **same characteristic** as the **first idle period**



Throughput analysis - part 1

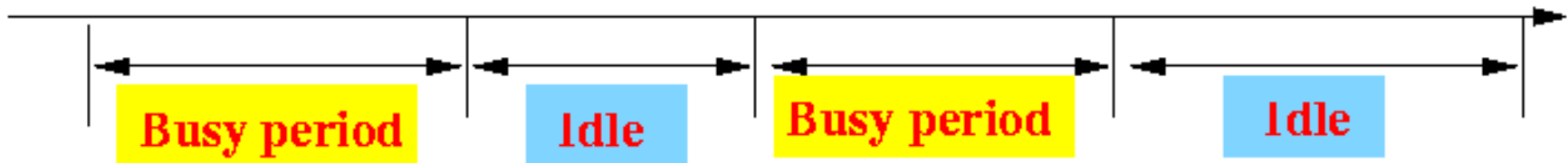
The following diagram gives the **factors** that determine the **throughput** of the **non-persistent CSMA** protocol:

-



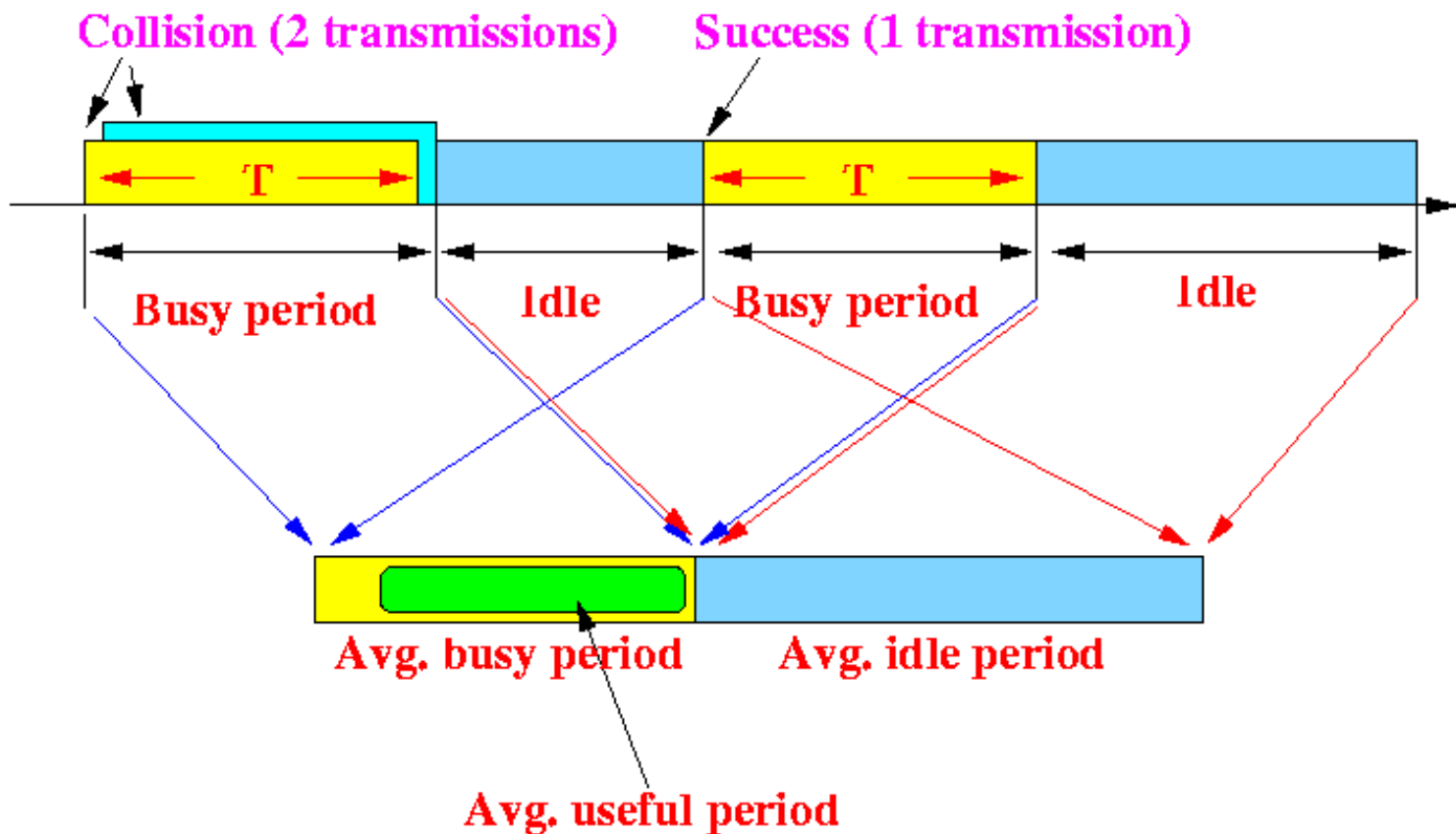
Observations:

- There is a **repeating pattern**:



- The **length** of a **busy period** is variable (random)
(depending on the **last starting transmission** in the **busy period**)
- The **length** of an **idle period** is variable (random)
(depending on the **time of the next transmission** that **starts** the subsequent **busy period**)

- **Abstraction:** represent the repeating pattern by one *average* pattern

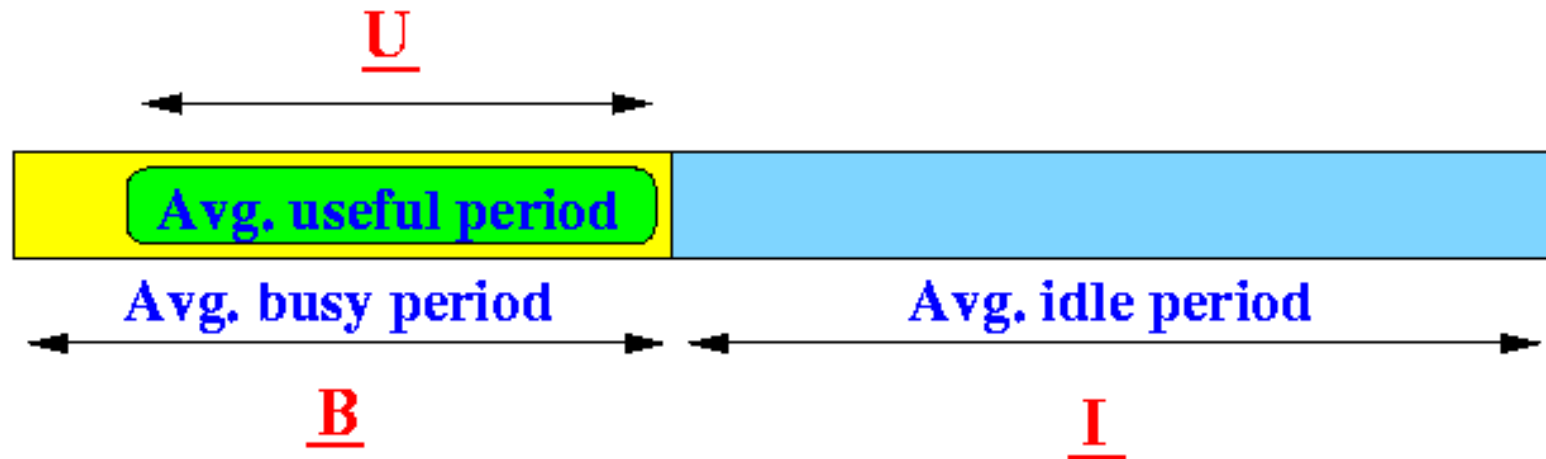


Definitions and notations:

- **$\underline{B}$** = the average duration of a busy period in the non-persistent CSMA cycle
- A busy period can be useful (i.e., successful transmission) or not useful (i.e., collision)
- **$\underline{U}$** = the average duration of a busy period that is useful in the non-persistent CSMA cycle
- **$\underline{I}$** = the average duration of a idle period in the non-persistent CSMA cycle



Throughput of the *unslotted non-persistent* CSMA protocol:

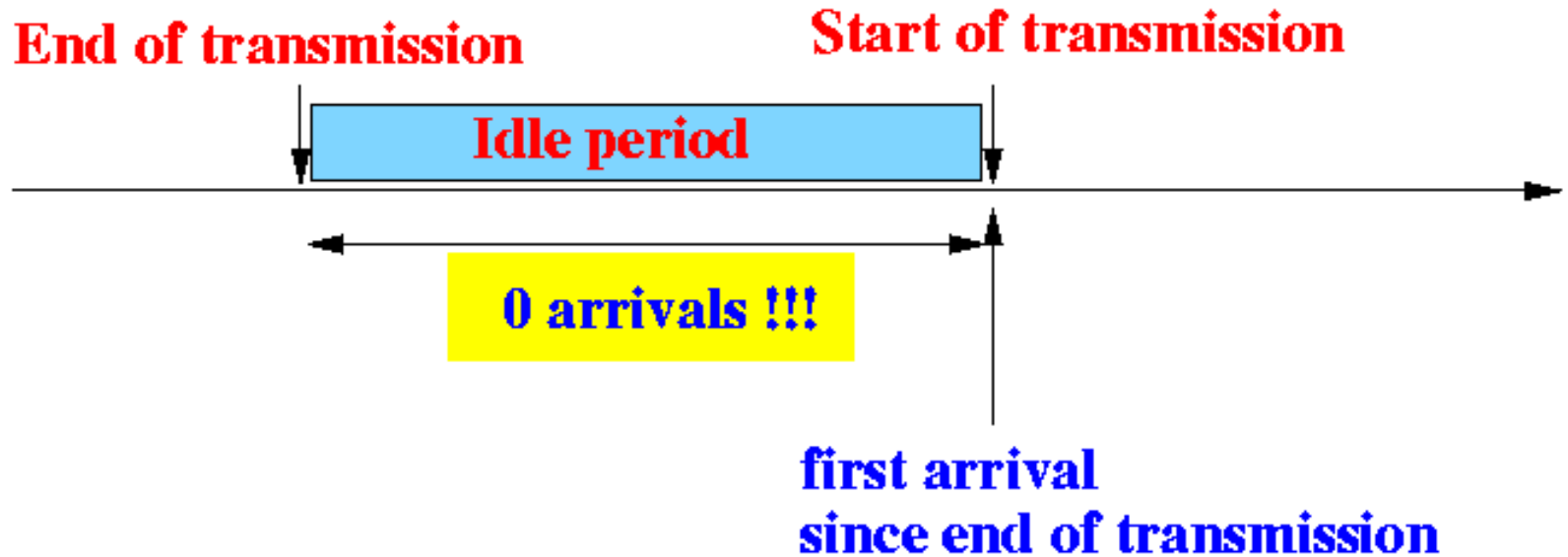


$$S_{\text{non-persistent CSMA}} = \frac{\underline{U}}{\underline{B} + \underline{I}} \dots\dots (1)$$

•All we need to do now is to find $\underline{U}$, $\underline{B}$, and $\underline{I}$

The average duration of an *idle period* (I)

- Characteristics of an idle period:
-



Notes:

- The **length (duration)** of an **idle period** is a **random variable**
- To **compute the *average* length** of an **idle period (random variable)** we must **first** find its **probability distribution function (pdf)**
- An **idle period** ends with the **first transmission**
- **Within the idle period** there are **0 message arrivals !**



The *cumulative* density function of an idle period I is therefore:

$$\begin{aligned} P[I \leq x] &= 1 - P[I > x] \\ &= 1 - P[\text{no packet arrives in } x \text{ sec}] \end{aligned}$$

$$P[k \text{ arrivals in } t \text{ sec}] = \frac{(gt)^k}{k!} e^{-gt}$$

$$P[I \leq x] = 1 - \frac{(gx)^0}{0!} e^{-gx}$$

$$\Leftrightarrow P[I \leq x] = 1 - e^{-gx} \dots \dots \dots (2)$$



The *probability* density function of an idle period I is then:

$$f_I(x) = \frac{d \mathbb{P}[I \leq x]}{d x}$$

$$\Leftrightarrow f_I(x) = \frac{d (1 - e^{-gx})}{d x}$$

$$\Leftrightarrow f_I(x) = g e^{-gx} \dots\dots (3)$$

Now we can **calculate the average duration of an idle period I**:

$$\underline{I} = \int_0^{\infty} x f_I(x) dx$$

(= the definition of expected value)

$$\Leftrightarrow \underline{I} = \int_0^{\infty} x g e^{-gx} dx$$

$$\Leftrightarrow \underline{I} = -x e^{-gx} - \frac{1}{g} e^{-gx} \Big|_{x=0}^{x=\infty}$$

$$\Leftrightarrow \underline{I} = (0 - 0) - \left(0 - \frac{1}{g} \right)$$

$$\Leftrightarrow \underline{I} = \frac{1}{g} \dots \dots (4)$$



One down, two more to go...



Average length of a useful period $\underline{U}$

- Recall that the length of each transmission is equal to T
- 2 kinds of busy periods:
- **Useful**
- A busy period that contains exactly *one* transmission is useful (no collision)
- The length (duration) of a busy period that contains exactly *one* transmission is equal to T (= length of one transmission)
- The contribution of such busy period to the total useful period is T



- **Not useful**

- A **busy period** that contains **more than *one* transmission** is **not useful (collision)**
- The **length (duration)** of a **busy period** that contains ***multiple* transmissions** is **random** (and difficult to compute)
- **However**, the **contribution** of such **busy period** to the **total useful period** is **0 (zero) !!!**



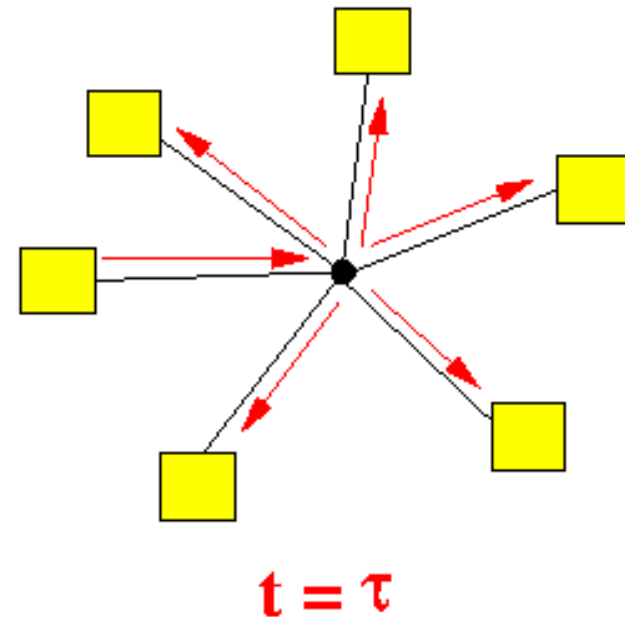
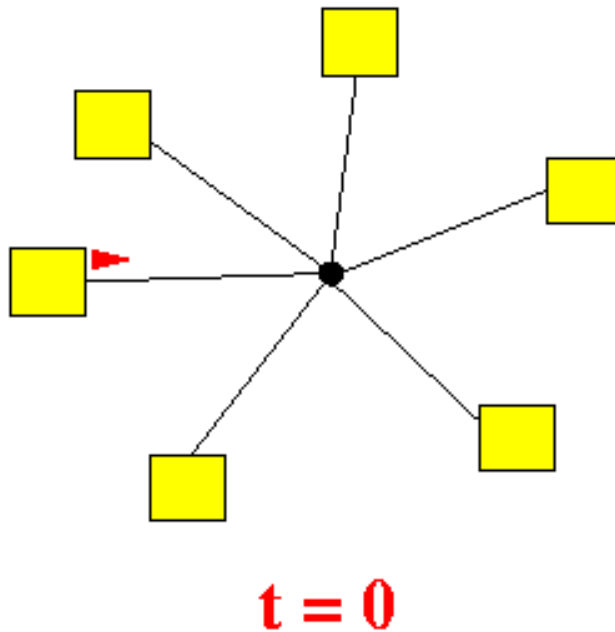
Average length of the useful period:

- The **average length** of the **useful periods** is the following **weight average**:
- $\underline{U} = T \times P[\text{transmission successful}] + 0 \times (1 - P[\text{transmission successful}])$

Note: We actually consider Propagation Time which is necessary to take the bits to the other nodes from the source is also an overhead, So only T is considered useful



A **busy period** is **successful**, i.e.,
contains **exactly *one*** transmission if and only if:

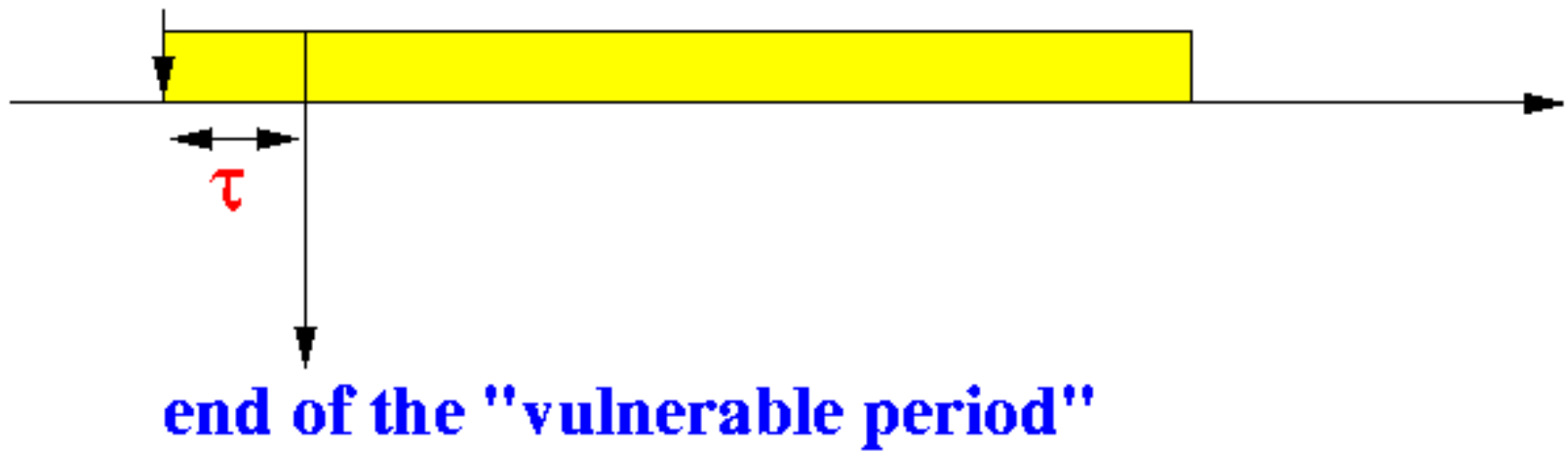


Some node starts transmitting

- Within τ sec of the start of this transmission, no *other* nodes will start a transmission
- Then after τ sec, when a node senses the transmission medium it will detect an *on-going transmission* and will refrain from starting its transmission
- Thus, $P[\text{transmission successful}]$ is equal to:
Probability of ZERO arrival in the vulnerability period



start of a transmission



$P[\text{ transmission successful }]$

$= P[0 \text{ arrivals within "vulnerable period" }]$

$= P[0 \text{ arrivals within } \tau \text{ sec }]$

$$P[k \text{ arrivals in } t \text{ sec }] = \frac{(gt)^k}{k!} e^{-gt}$$

$$P[\text{transmission successful}] = \frac{(g\tau)^0}{0!} e^{-g\tau}$$

$$\Leftrightarrow P[\text{transmission successful}] = e^{-g\tau}$$

..... (5)

Therefore:

$$\underline{U} = T \times P[\text{transmission successful}]$$

$$+ 0 \times (1 - P[\text{transmission successful}])$$

$$= T \times e^{-g\tau} + 0 \times (1 - e^{-g\tau})$$

Let us think of this also as a Random Variable with two possible values T or 0.

So average value is $[T \times \text{chance of } T] + [0 \times \text{Chance of } 0]$

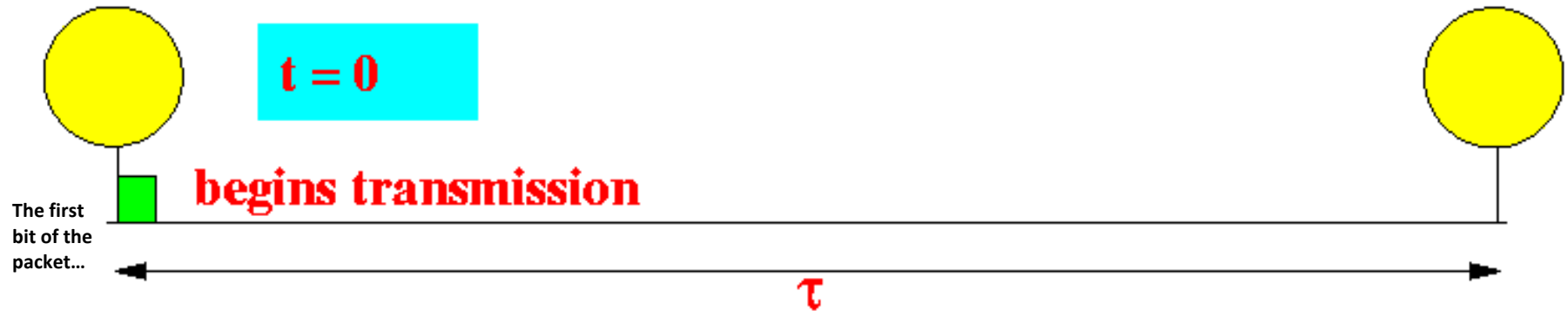


- **Two down, one more to go....**

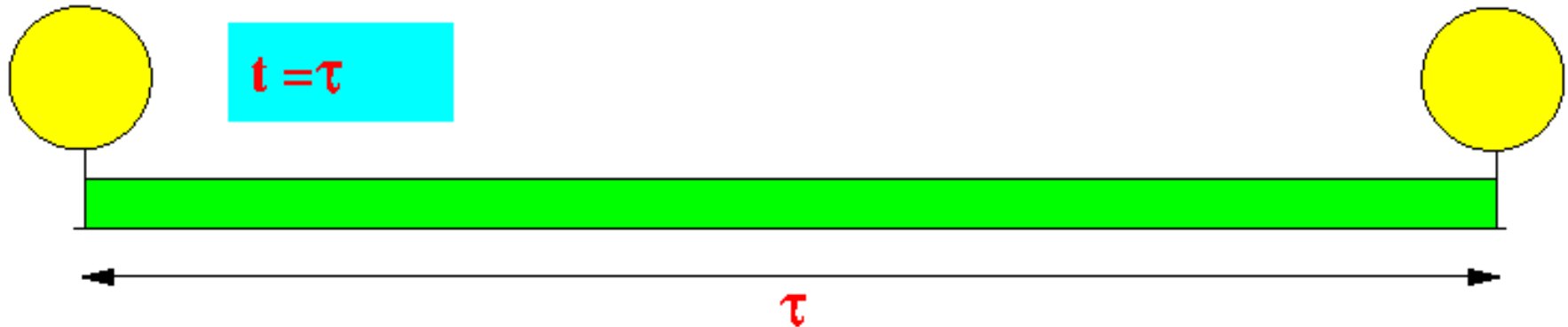
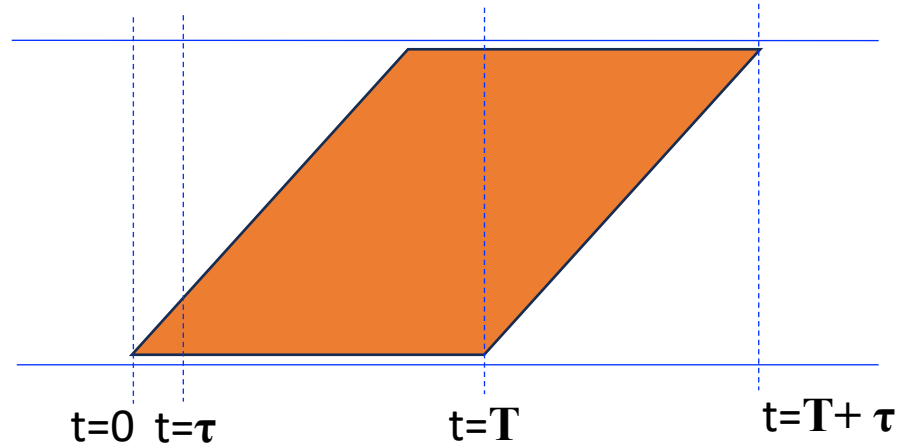
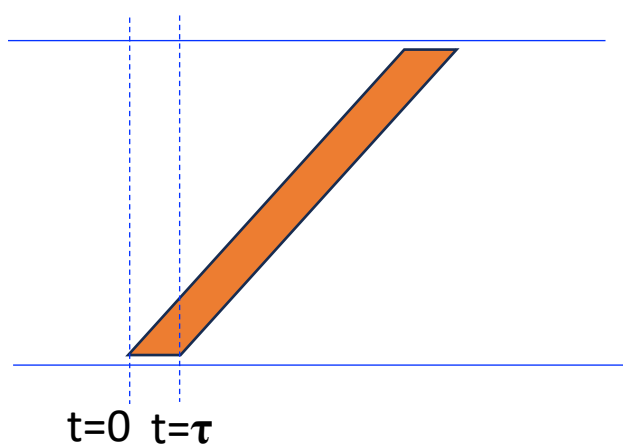


Average length of a busy period B - part 1

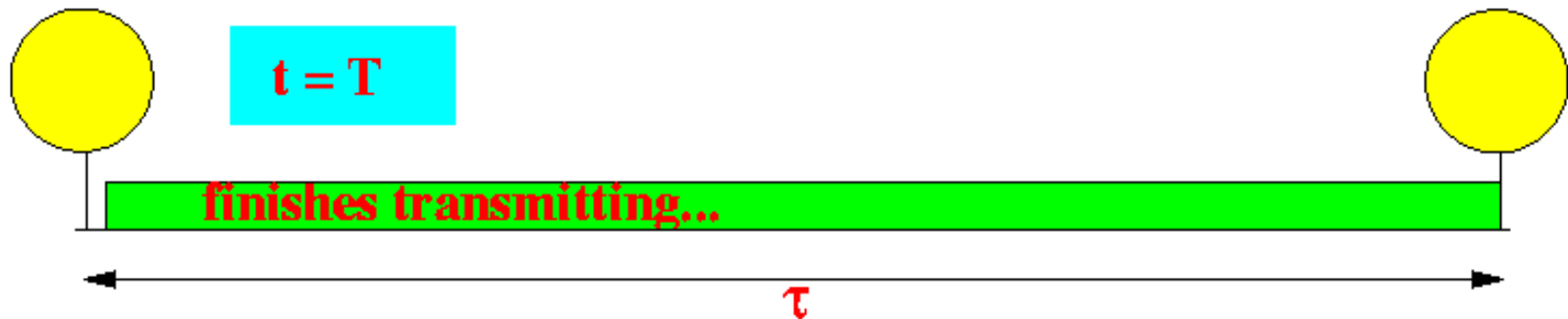
- *Anatomy* of a (simple) busy period:
- A busy period begins with some node starting a transmission:



The transmission reaches the end of the cable at $t = \tau$:



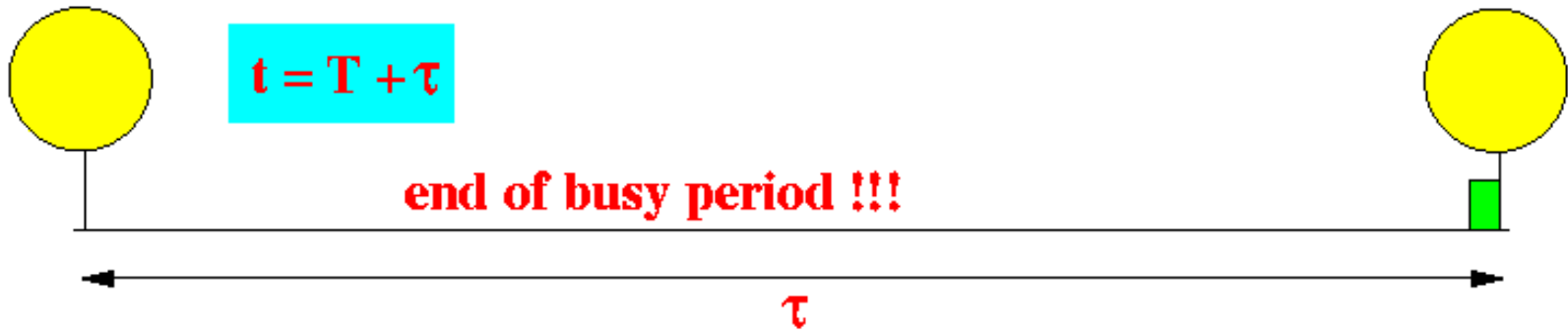
- The node finishes transmitting after $t = T$ sec:



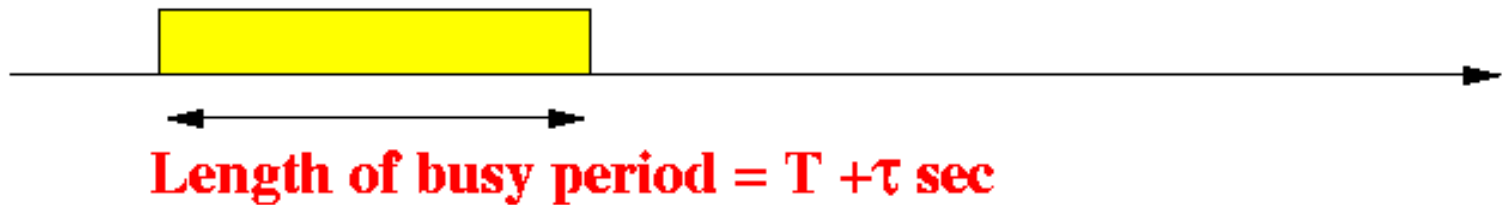
Notice that the **transmission medium is not yet idle...**

I.e., the **busy period has *not* yet ended !!!**

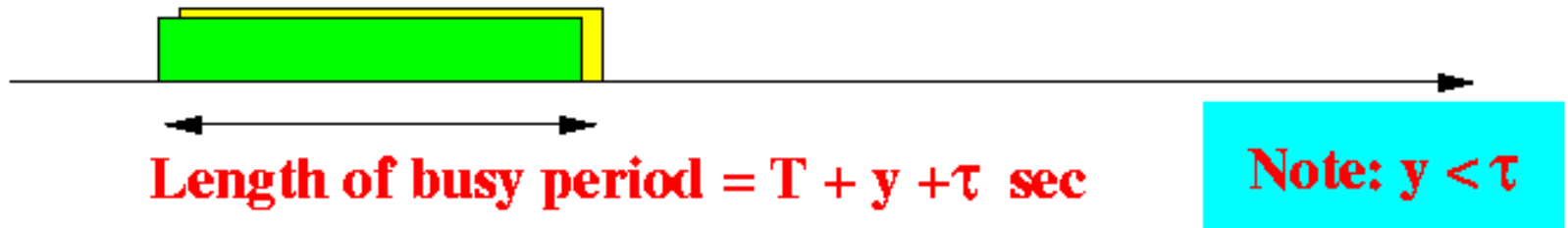
- The **transmission clears the cable** at $t = T + \tau$:



- The length of a busy period is a random variable due to possible nearly-*simultaneous* starting transmissions:
- One arrival:



Multiple (nearly simultaneous) arrivals:



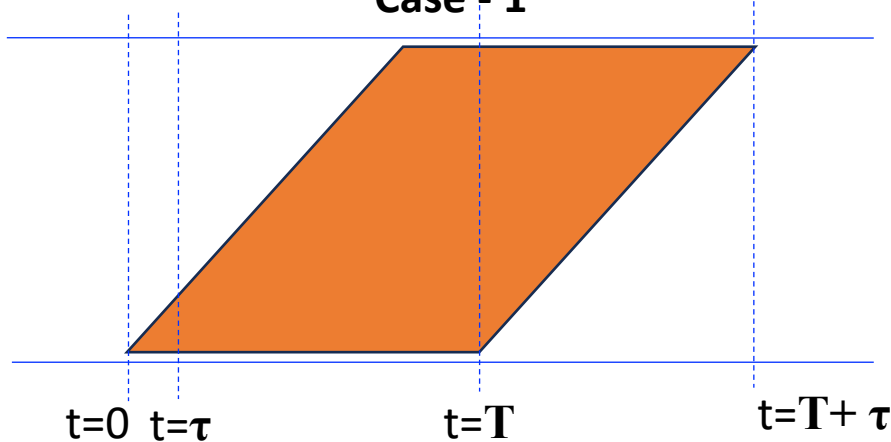
Notes:

- If there is **one transmission** in the **busy period**, the **length of the busy period** will be equal to $(T + \tau)$ sec
- But when there are **multiple (nearly simultaneous) transmissions**, the **length of the busy period** will be equal to $(T + y + \tau)$ sec, where:
- y = the **time lag** between the **starting transmission** and the **last transmission** in the **busy period**



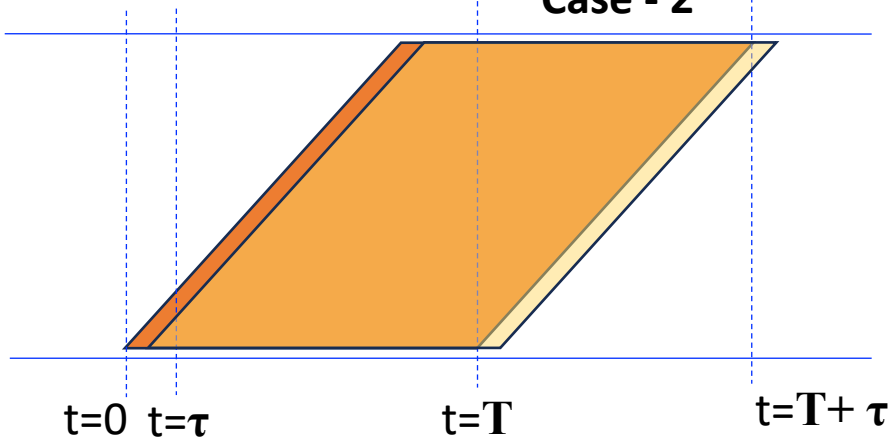
Cases -

Case - 1



No collision, time = $t = T + \tau$

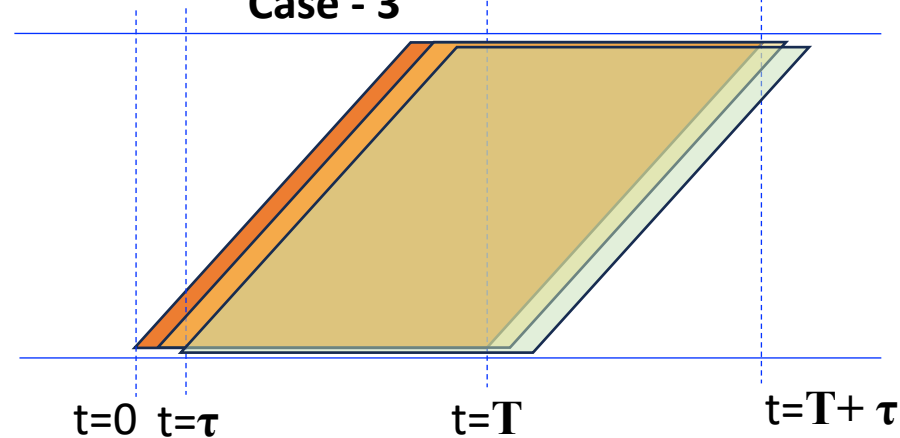
Case - 2



Collision – Another packet starts at $t = \tau/2$

Total time = $T + \tau + \tau/2$

Case - 3



Collision – Another packet starts at $t = \tau - \Delta$

Total time = $T + \tau + \tau - \Delta$

Average length of a busy period B

- From the discussion above: $B = T + \tau + y \dots\dots (7)$
- where:
- T = length of a packet transmission (constant)
- τ = (max) end-to-end delay (constant)
- y = time lag of the last transmission (random)
- Here y is the Random variable which can be 0 to τ



From the theory of probability...

$$E[B] = E[T + \tau + y]$$

$$\Leftrightarrow E[B] = T + \tau + E[y]$$

$$\Leftrightarrow \underline{B} = T + \tau + y \quad \text{..... (8)}$$

- In order to compute the **expected value $E[y]$ ($\underline{y}$)**,
- We need to find the **pdf**
- **(Probability distribution function)** of y first.



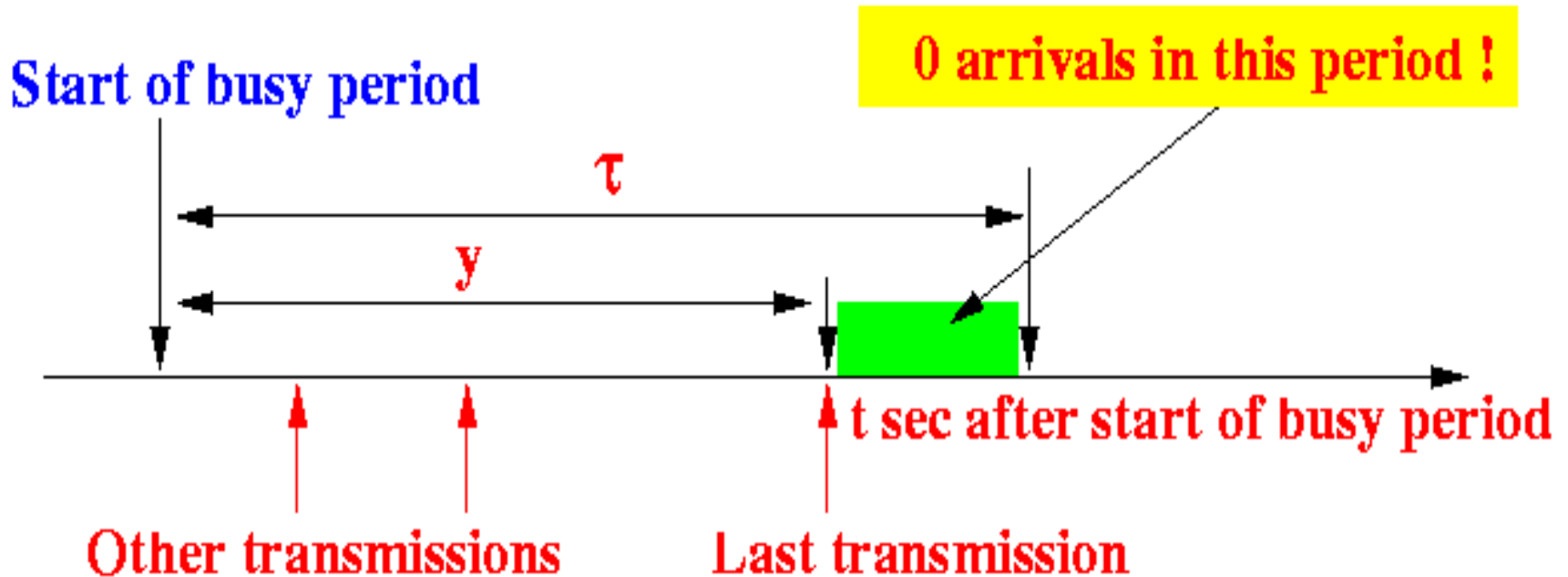
The pdf of the random variable y

- The random variable y is the lag time of the last transmission in the busy period.
- Since the propagation delay is at most τ , the lag time of any transmission is at most τ
- Because after τ sec, a node that wants to *transmit* will detect that the channel is *busy* and will refrain from transmitting



Situation

So think of the possible values of y , and generalize it by $y \leq t$



Therefore:

$$P[y \leq t] = P[0 \text{ arrivals in } (\tau - t) \text{ sec}]$$

$$P[k \text{ arrivals in } t \text{ sec}] = \frac{(gt)^k}{k!} e^{-gt}$$

Therefore:

$$P[y \leq t] = \frac{(g(\tau-t))^0}{0!} e^{-g(\tau-t)} \dots\dots (9)$$

$$\Leftrightarrow P[y \leq t] = e^{-g(\tau-t)} \quad (\text{with } t \in [0, \tau))$$

We can now obtain the **probability density function (pdf) f_y**

$$f_y(t) = \frac{d \mathbb{P}[y \leq t]}{d t}$$

$$\Leftrightarrow f_y(t) = \frac{d e^{-g(\tau-t)}}{d t} \quad (\text{Use "chain rule"})$$

$$\Leftrightarrow f_y(t) = e^{-g(\tau-t)} \times (-g) \times (-1)$$

$$\Leftrightarrow f_y(t) = g e^{-g(\tau-t)} \quad (\text{with } t \in [0, \tau))$$

..... (10)

Problem with the pdf that we found....

- Every pdf $f(t)$ must satisfy the following property:

$$-\infty \int^{\infty} f(t) dt = 1$$

..... (11)

- (because the **cumulative outcome** of *every possible* value is equal to **certainty** (1))
- This **property** does **not hold** for $f_y(t)$ in the **equation (10)**:



Checking of $f_y(t)$ is a valid PDF

$$\begin{aligned} \int_{-\infty}^{\infty} f_y(t) dt &= \int_{-\infty}^{\infty} g e^{-g(\tau-t)} dt \\ &= \int_0^{\tau} g e^{-g(\tau-t)} dt \\ &= e^{-g(\tau-t)} \Big|_{t=0}^{t=\tau} \\ &= e^{-g(\tau-\tau)} - e^{-g(\tau-0)} \\ &= 1 - e^{-g\tau} \\ &\neq 1 !!! \end{aligned}$$



Reason of this behavior:

- y is a hybrid random variable:
- $f_y(t)$ is continuous on $(0, \tau)$, but...
- $f_y(t)$ is discontinuous in $t = 0$...
- In fact, in $t = 0$, the random variable y behaves like a discrete random variable:

The missing value – Explore -

$$\begin{aligned}P[y = 0] &= P[\text{no "last arriving" transmission}] \\&= P[\text{one transmission in busy period}] \\&= P[\text{successful transmission}]\end{aligned}$$

$$= e^{-g\tau}$$

- The value $e^{-g\tau}$ is equal to the "missing value" in the property above !!!

The correct pdf of the random variable y :

$$f_y(t) = e^{-g\tau} \times \delta(t) + g e^{-g(\tau-t)}$$

(with $t \in [0, \tau)$)

The function $\delta(t)$ is defined as: (11)

$\delta(t) = 1$ if $t = 0$

$\delta(t) = 0$ otherwise



The expected value of the random variable y

- We can now compute the **expected value** $E[y]$:
-

$$\begin{aligned} E[y] &= \int_{-\infty}^{\infty} t \times f_y(t) dt \\ &= \int_0^{\tau} t \times f_y(t) dt \end{aligned}$$

Split off the **discontinuous point** $t = 0$

$$\begin{aligned}
 E[y] &= 0 \times P[y = 0] + \int_0^\tau t \times f_y(t) dt \\
 &= 0 \times e^{-g\tau} + \int_0^\tau t \times g e^{-g(\tau-t)} dt \\
 &= g e^{-g(\tau)} \int_0^\tau t \times e^{-g(-t)} dt \\
 &= g e^{-g\tau} \int_0^\tau t \times e^{gt} dt
 \end{aligned}$$

$$\Leftrightarrow E[y] = g e^{-g\tau} \frac{1 - e^{g\tau} + g\tau e^{g\tau}}{g^2}$$

$$= e^{-g\tau} \frac{1 - e^{g\tau} + g\tau e^{g\tau}}{g}$$

$$= \frac{e^{-g\tau} - 1 + g\tau}{g}$$

$$= \tau - \frac{1 - e^{-g\tau}}{g}$$

..... (12)

Average length of a busy period $\underline{B}$ - part 2

- Recall that the average length of a busy period $\underline{B}$ was
- $\underline{B} = T + \tau + y \dots\dots (8)$
- With Equation (12), we can now complete the derivation:

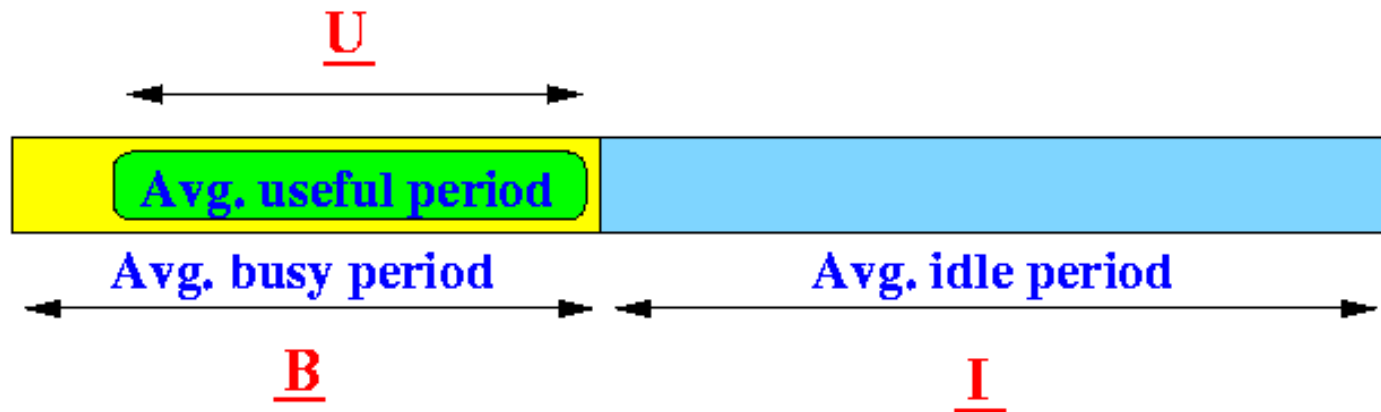
$$\underline{B} = T + \tau + \left(\tau - \frac{1 - e^{-g\tau}}{g} \right)$$

$$= T + 2\tau - \frac{1 - e^{-g\tau}}{g}$$

$\dots\dots (13)$

Throughput analysis of unslotted non-persistent CSMA - part 2

- Recall that: throughput of the *unslotted non-persistent* CSMA protocol:
-



$$S_{\text{non-persistent CSMA}} = \frac{\underline{U}}{\underline{B} + \underline{I}} \dots\dots (1)$$

We have found that:

$$\underline{I} = \frac{1}{g} \dots\dots (4)$$

$$\underline{U} = T \times e^{-g\tau} \dots\dots (6)$$

$$\underline{B} = T + 2\tau - \frac{1 - e^{-g\tau}}{g} \dots\dots (13)$$

$$S = \frac{\underline{U}}{\underline{B} + \underline{I}}$$

($\times g/g$)

$$\Leftrightarrow S = \frac{T \times e^{-g\tau}}{T + 2\tau - (1 - e^{-g\tau})/g + 1/g}$$

$$\Leftrightarrow S = \frac{g T \times e^{-g\tau}}{g(T + 2\tau) - (1 - e^{-g\tau}) + 1}$$

($-1 + 1 = 0$)

$$\Leftrightarrow S = \frac{g T \times e^{-g\tau}}{g(T + 2\tau) + e^{-g\tau}}$$

(sub: $a = \tau/T$, or $\tau = aT$)

$$\Leftrightarrow S = \frac{g T \times e^{-gaT}}{g(T + 2aT) + e^{-gaT}}$$

(move gT out)

$$\Leftrightarrow S = \frac{gT \times e^{-agT}}{gT(1 + 2a) + e^{-agT}}$$

..... (14)

Recall that:

- **Offered load (rate) g = number of (old and new) messages that contend for the channel per time unit**
- **Define:**
- **$G = gT$**
- **G = number of arrivals during one packet transmission**
-

We can write the **throughput formula (14)** a bit nicer as:

$$S = \frac{G \times e^{-aG}}{G(1 + 2a) + e^{-aG}}$$

..... (15)

Performance graphs:

$a = \tau/T$

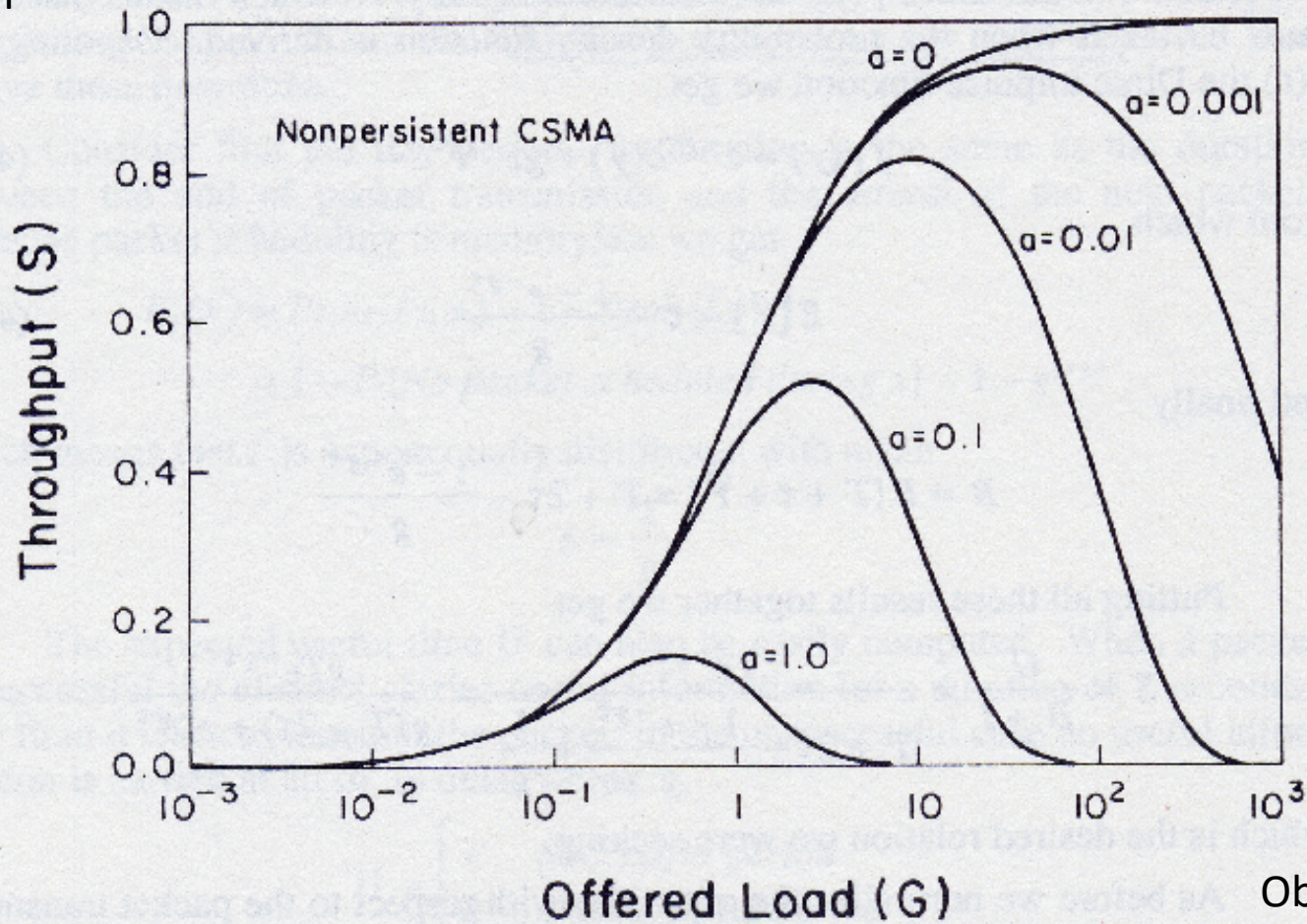


Figure 4.2: Throughput-Load of Nonpersistent CSMA

Observe the effect of a

